

Local Economic Spillovers from Water Market Reform

Billy Ferguson

Zane Kashner

UC Berkeley ARE

Stanford GSB

February 24, 2026

Motivation



- **Murray-Darling Basin:** Largest basin in Australia.
- Productive agriculture is irrigation dependent.
- Globally, $< 1\%$ of water rights are traded.

Motivation



- **Murray-Darling Basin:** Largest basin in Australia.
- Productive agriculture is irrigation dependent.
- Globally, $< 1\%$ of water rights are traded.
- 2007 Reform $\Rightarrow$ **most active and valuable** market.
 - Trade volumes greater than 25%
 - 60% of farms have traded.
 - **\$2.3 billion** gains from 2007-2015 (Rafey 2023).

Motivation



- **Murray-Darling Basin:** Largest basin in Australia.
- Productive agriculture is irrigation dependent.
- Globally, $< 1\%$ of water rights are traded.
- 2007 Reform $\Rightarrow$ **most active and valuable** market.
 - Trade volumes greater than 25%
 - 60% of farms have traded.
 - **\$2.3 billion** gains from 2007-2015 (Rafey 2023).
 - *Crown Jewel of Water Marketing*

Motivation



- **Murray-Darling Basin:** Largest basin in Australia.
- Productive agriculture is irrigation dependent.
- Globally, $< 1\%$ of water rights are traded.
- 2007 Reform $\Rightarrow$ **most active and valuable** market.
 - Trade volumes greater than 25%
 - 60% of farms have traded.
 - **\$2.3 billion** gains from 2007-2015 (Rafey 2023).
 - *Crown Jewel of Water Marketing*
- Motivates market-based agenda

Motivation



- **Murray-Darling Basin:** Largest basin in Australia.
- Productive agriculture is irrigation dependent.
- Globally, $< 1\%$ of water rights are traded.
- 2007 Reform $\Rightarrow$ **most active and valuable** market.
 - Trade volumes greater than 25%
 - 60% of farms have traded.
 - **\$2.3 billion** gains from 2007-2015 (Rafey 2023).
 - *Crown Jewel of Water Marketing*
- Motivates market-based agenda...among economists.

Motivation

Water trading's 'unintended' consequences across Australia's southern Murray-Darling Basin By national rural reporter Kath Sullivan Fri 12 Jul 2019

Fri 12 Jul 2019



Men burn copies of the Murray-Darling Basin guide. (Gabrielle Dunlevy: AAP)



Motivation

- Water markets improve efficiency (Rafey 2023)
- Local harms in selling communities (Wheeler et al. 2024)



Motivation

- Water markets improve efficiency (Rafey 2023)
- Local harms in selling communities (Wheeler et al. 2024)

"Because of permanent trade, people walked off and left the district. The population decreased, income went down with water taken off the land. It flattens everyone. You drive past a beautiful thriving farm and see it just let go. It upsets people"

Farmer from Goulburn Valley, Victoria
(Hamilton and Kells 2021)



Motivation

- Water markets improve efficiency (Rafey 2023)
- Local harms in selling communities (Wheeler et al. 2024)

"I'm not sure the Imperial Valley is a big believer in water markets," Shields said. "We're a big believer in protecting our community."

Manager of Imperial Irrigation District, CA
(Smolens 2023)



Motivation

- Water markets improve efficiency (Rafey 2023)
- Local harms in selling communities (Wheeler et al. 2024)

"From a community point of view, perhaps we need to consider maintaining water in the regions, but then as a business person would I like to be restricted to sell my water only within my region when I might be able to earn double for it downstream? It's about having your cake and eating it too."

Business Chamber VP from Pretty Pine, NSW
(Sullivan 2019)



This Paper

“From a community point of view, perhaps we need to consider maintaining water in the regions, but then as a business person would I like to be restricted to sell my water only within my region when I might be able to earn double for it downstream? It’s about having your cake and eating it too” (Sullivan 2019)

Research Questions

1. What are the **local economic spillovers** from water trade?
2. How large are community impacts relative to **gains from trade**?
3. How do we better **design markets** with this in mind?

Preview of Results

- **Local spillovers:** At the median local loss of water for irrigation, we see:
 - 7% decrease in wages and 15% decrease in agricultural employment.
 - 12% decrease in population and 16% decrease in home prices.
 - Document political response and increased deaths of despair.
- **Community value:** Household residential choice preferences for local irrigation yield:
 - Median community willingness-to-pay for water of **\$49/ML**.
 - Large variance with 25% of communities having values over **\$180/ML**.
- **Farmer value:** Agricultural production function estimation yields:
 - Median farmer willingness-to-pay of **\$247/ML**.
- **Market design:**
 - Farmer and community willingness-to-pay are **negatively correlated**.
 - There existed sufficient surplus to achieve Pareto gains.
 - Without transfers from buying community's, Pareto improvements fail for **43%** of trades.

Contributions

- **Water Market Design:** Akhundjanov et al. (2025), Sum (2025), Ferguson (2025), Ferguson and Milgrom (2024), Hagerty (2025), Rafey (2023), Regnacq et al. (2016), Colby (1990), Hanak (1990).
 - Study key bottleneck to policy reform using world's largest, most active water market.
 - First market-wide analysis of distributional impacts.
- **Distributional Consequences of Trade:** Sum (2025), Apselund (2025), Calliendo et al. (2019), Fajgelbaum and Khandelwal (2014), Autor et al. (2013), Kovak (2013).
 - Environmental input market where distribution is first-order.
 - Integrate event-study, migration model, and production to analyze tradeoffs.
- **Place-Based Policy:** Slattery (2025), Kashner (2025), Kline and Moretti (2013), Glaeser and Gottlieb (2008).
 - Place-based policy under exchange where losers are more pronounced.

Roadmap

1. Water in Australia
2. Conceptual Framework
3. Communal Value of Water: Residential Choice
4. Market Value of Water: Agricultural Production
5. Market Design and Counterfactuals

Roadmap

1. Water in Australia
2. Conceptual Framework
3. Communal Value of Water: Residential Choice
4. Market Value of Water: Agricultural Production
5. Market Design and Counterfactuals

Water in Australia: Murray-Darling Basin



- Largest basin and most productive agriculture.
- Dry growing season $\Rightarrow$ irrigation dependent.
- Historically, farmer rights mostly fixed locally.
 - Strong local irrigation district control.
- Millenium Drought $\Rightarrow$ **Water Act of 2007**
 - Homogenized water rights.
 - Removed trade barriers and district restrictions.
- Farmers could respond to private incentives.

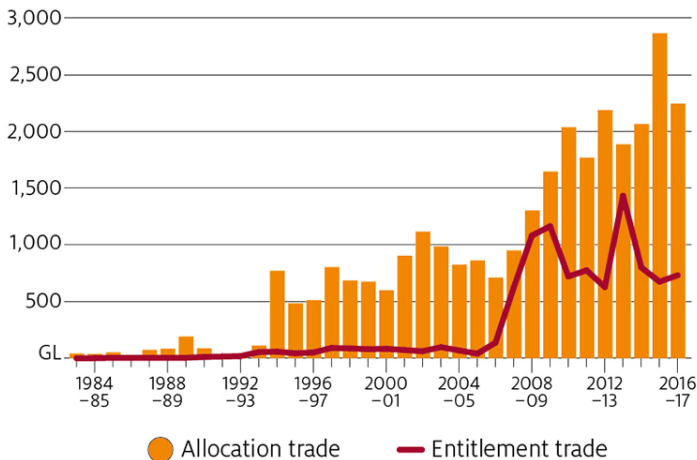
Water in Australia: Murray-Darling Basin



- Largest basin and most productive agriculture.
- Dry growing season $\Rightarrow$ irrigation dependent.
- Historically, farmer rights mostly fixed locally.
 - Strong local irrigation district control.
- Millenium Drought $\Rightarrow$ **Water Act of 2007**
 - Homogenized water rights.
 - Removed trade barriers and district restrictions.
- Farmers could respond to private incentives.
- **Most active and valuable** water market.
 - 60% of farms have traded.
 - **\$2.3 billion gains** from 2007-2015 (Rafey 2023).

Use Water Act as Natural Experiment

- **Ideal Experiment:**
Random permanent change
is water rights.
- **Natural Experiment:**
Water Act of 2007's
liberalization of trade $\Rightarrow$
 $\approx 20\times$ increase in
permanent transfers.
- **Intuition:** Persistent factor
misallocation unlocked by
policy.



Conceptual Framework



$$V_S(W_S)$$



$$V_B(W_B)$$

Conceptual Framework



$$\sum U_{S,h}(W_S)$$



$$V_S(W_S)$$



$$\sum U_{B,h}(W_B)$$



$$V_B(W_B)$$

Conceptual Framework



$$\sum U_{S,h}(W_S)$$



$$V_S(W_S)$$

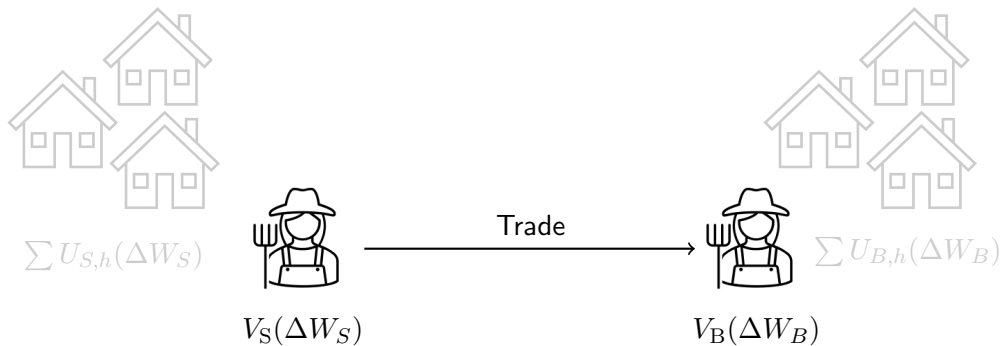


$$\sum U_{B,h}(W_B)$$

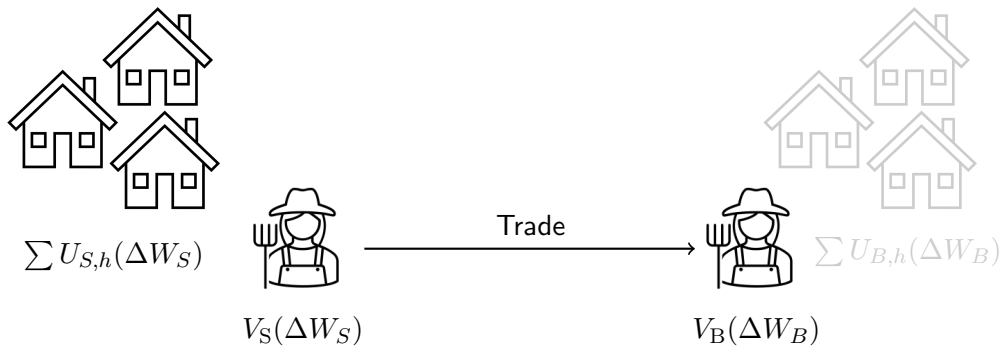


$$V_B(W_B)$$

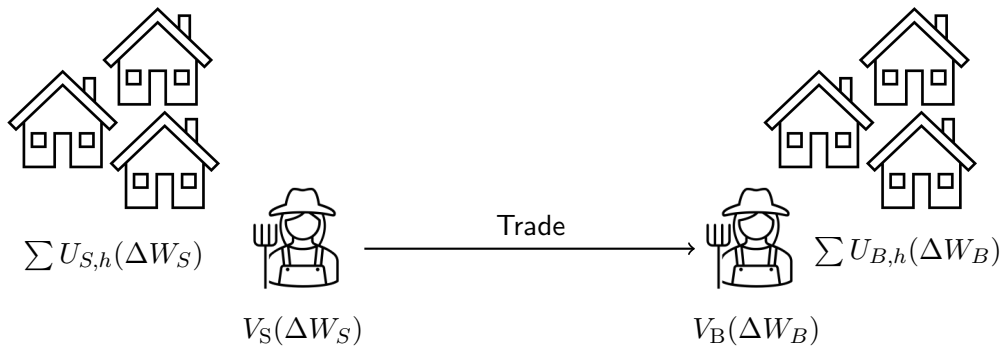
Conceptual Framework



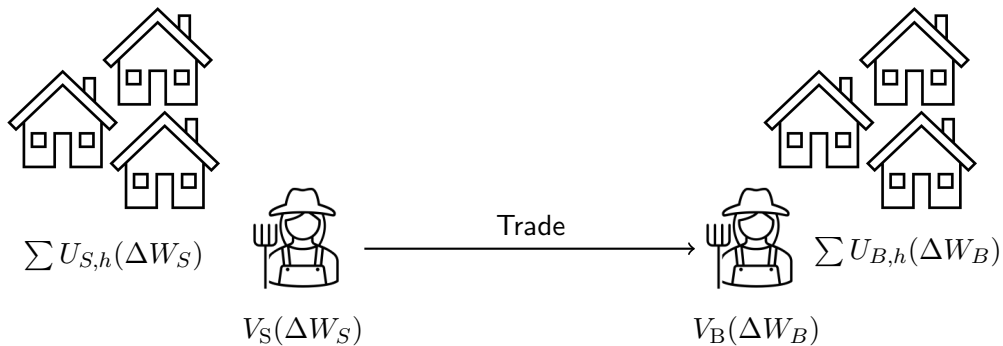
Conceptual Framework



Conceptual Framework

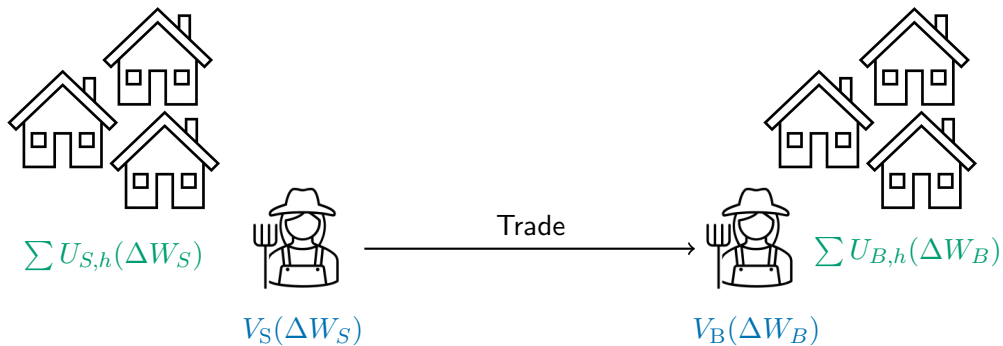


Conceptual Framework



Acheive Pareto gains?

Conceptual Framework



Acheive Pareto gains?

Roadmap

1. Communal Value of Water: Residential Choice

- **Goal:** Estimate household willingness-to-pay for local irrigation.

2. Market Value of Water: Agricultural Production

- **Goal:** Estimate private farmer WTP for water.

3. Market Design and Counterfactuals

- **Goal:** Compare community and private WTP to motivate market designs with Pareto gains.

Roadmap

1. Communal Value of Water: Residential Choice

- **Goal:** Estimate household willingness-to-pay for local irrigation.
- Data and identifying variation
- Motivate preference for local irrigation by documenting spillovers with event studies.
- Build and estimate model of residential choice with preference for local irrigation.

2. Market Value of Water: Agricultural Production

- Goal: Estimate private farmer WTP for water.

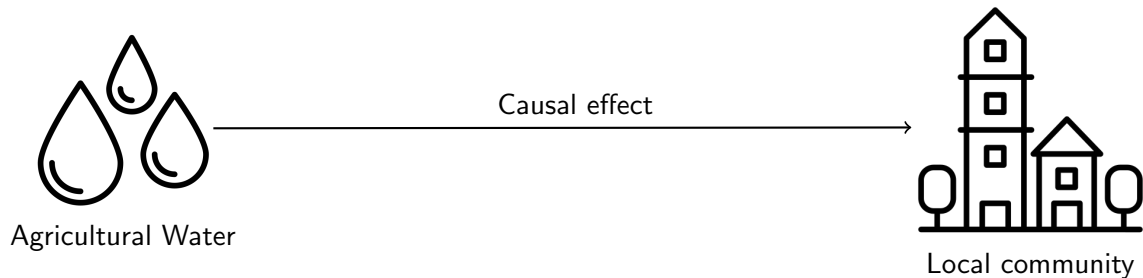
3. Market Design and Counterfactuals

- Goal: Compare community and private WTP to motivate market designs with Pareto gains.

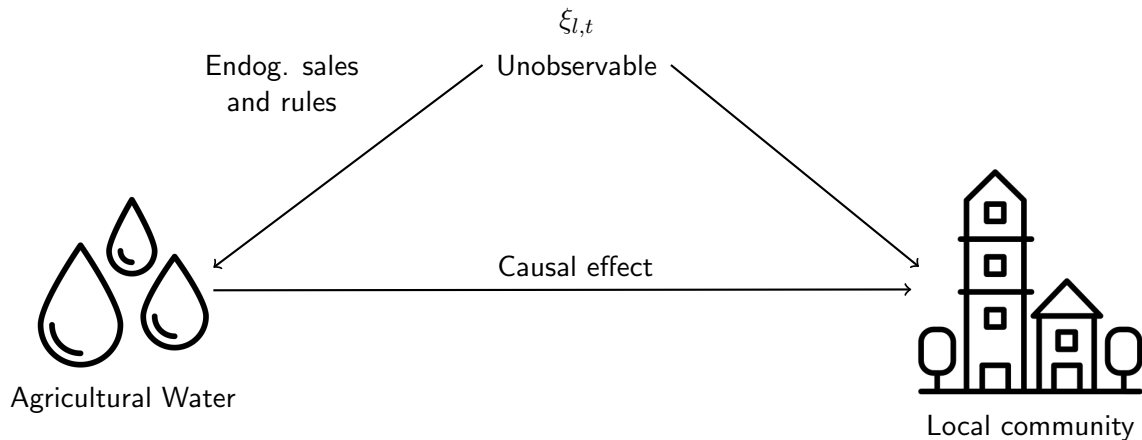
Data

- **Water usage:** SA1 $\times$ Year (2001-2020)
 - Australia's Terrestrial Ecosystem Research Network (TERN)
 - Combine applied ET measures from Sentinel-2 and Landsat satellite data
 - Measures average evapotranspiration (30m-level) in mm/day
- **Land Characteristics:** SA1 in 2001
 - Soil and Landscape Grid of Australia at 90m pixel (observed and predicted with ML)
 - Nonanthropogenic soil chars. for 161k points: Available Water Capacity (AWC) and depth
- **Census every 5 years:** SA1 $\times$ Year (1996-2021)
 - Population, rent, mortgage, wages, jobs.
 - Migration flows between region.
- **Property Sales:** Property $\times$ Transactions (1995-2024)
 - All transactions and parcels in NSW ($\approx$ 1/3 MDB's population)
- **Other:** voting by poll station, mortality causes, deaths by location, and rain

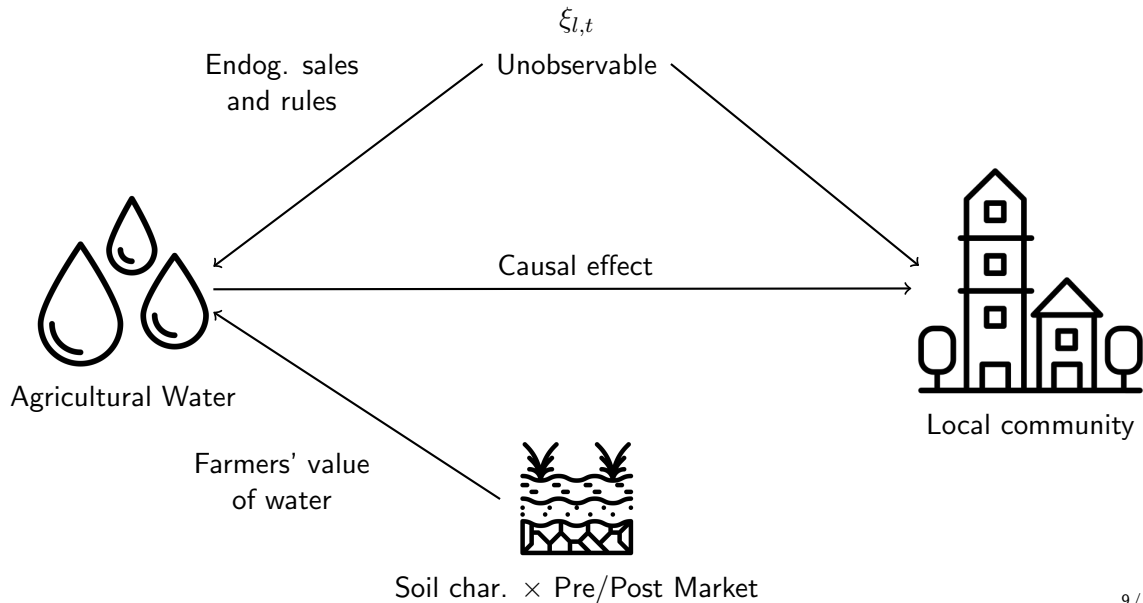
Documenting Local Spillovers: Research Design Intuition



Documenting Local Spillovers: Research Design Intuition



Documenting Local Spillovers: Research Design Intuition



Defining the water variable

- No local water rights/trade data, but granular evapotranspiration satellite data.

Defining the water variable

- No local water rights/trade data, but granular evapotranspiration satellite data.
- **Problem:** Evapotranspiration is effected by seasonal weather and rain.

Defining the water variable

- No local water rights/trade data, but granular evapotranspiration satellite data.
- **Problem:** Evapotranspiration is effected by seasonal weather and rain.
- **Solution:** Define SA1 $\times$ Month-Year residualized ET variable to capture available water.

$$ET_{l,m,t} = \beta Rain_{l,m,t} + \theta_m + \gamma_l + \varepsilon_{l,m,t} \qquad W_{l,t} = \sum_{m \in Dry} \hat{ET}_{imt}^{resid}$$

- Define annual SA1 residual ET by only summing over dry months (Oct-Mar.)

Defining the water variable

- No local water rights/trade data, but granular evapotranspiration satellite data.
- **Problem:** Evapotranspiration is effected by seasonal weather and rain.
- **Solution:** Define SA1 $\times$ Month-Year residualized ET variable to capture available water.

$$ET_{l,m,t} = \beta Rain_{l,m,t} + \theta_m + \gamma_l + \varepsilon_{l,m,t}$$

$$W_{l,t} = \sum_{m \in Dry} \hat{ET}_{imt}^{resid}$$

- Define annual SA1 residual ET by only summing over dry months (Oct-Mar.)
 - Dry months capture variation in regional allocation and trade data. [ResidCompare](#)

Defining the water variable

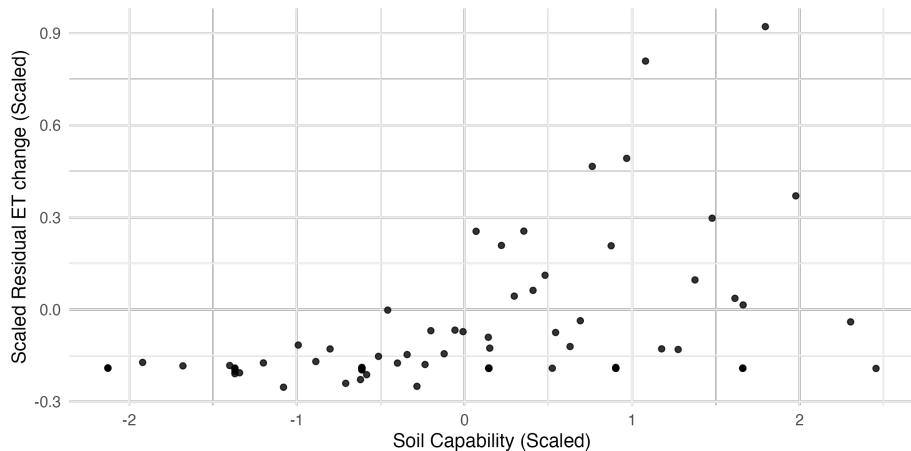
- No local water rights/trade data, but granular evapotranspiration satellite data.
- **Problem:** Evapotranspiration is effected by seasonal weather and rain.
- **Solution:** Define SA1 $\times$ Month-Year residualized ET variable to capture available water.

$$ET_{l,m,t} = \beta Rain_{l,m,t} + \theta_m + \gamma_l + \varepsilon_{l,m,t} \qquad W_{l,t} = \sum_{m \in Dry} \hat{ET}_{imt}^{resid}$$

- Define annual SA1 residual ET by only summing over dry months (Oct-Mar.)
 - Dry months capture variation in regional allocation and trade data. [ResidCompare](#)
- Water change is the difference between pre/post-2007 average dry month residualized ET.

$$\Delta W_l = \overline{W}_{l,post-2007} - \overline{W}_{l,pre-2007}$$

Ex-ante factor misallocation: Water change and soil quality FirstStage



Note: Soil quality is from a harmonized land capability mapping in [Adams & Engert 2023]. Average change in SA1 evapotranspiration (ML/hectare). Distribution Geography SoilGeography

DiD IV approach

$$\log(Y_{l,t}) - \log(Y_{l,06}) = \tau_t \Delta \hat{W}_l + \phi_{d,t} + \varepsilon_{l,t}$$

$Y_{l,t}$: Outcomes in location l in year t (household income, rent, mortgage, and population)

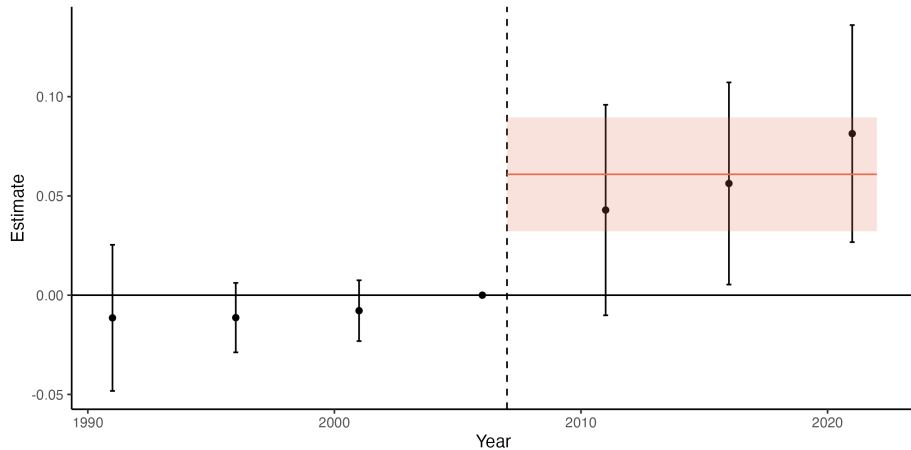
$\Delta \hat{W}_l$: Average difference in residual evapotranspiration per hectare

IV w/ natural soil characteristics: Available Water Capacity (AWC), Depth, and pH.

τ_t : Effect of change in water

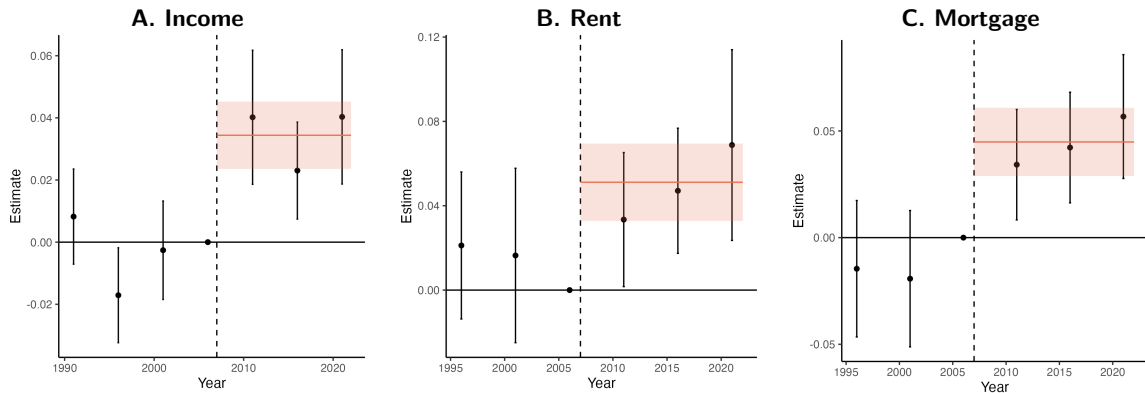
$\phi_{d,t}$: Water-district fixed effect

Populations respond more slowly to shift in water



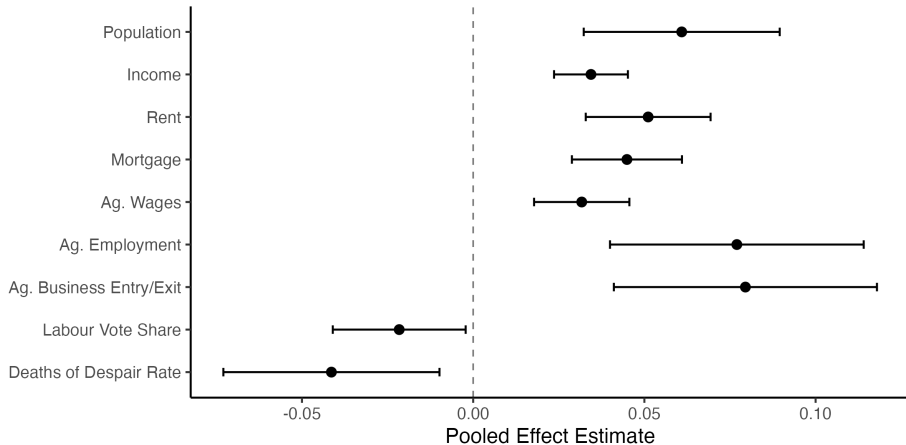
Notes: Effect of permanent increase of 0.0816 ET/hectare on SA1 population. Equivalent to a 10% increase in irrigation intensity at the average SA1. [Table](#)

Event Study Results for Local Economic Indicators

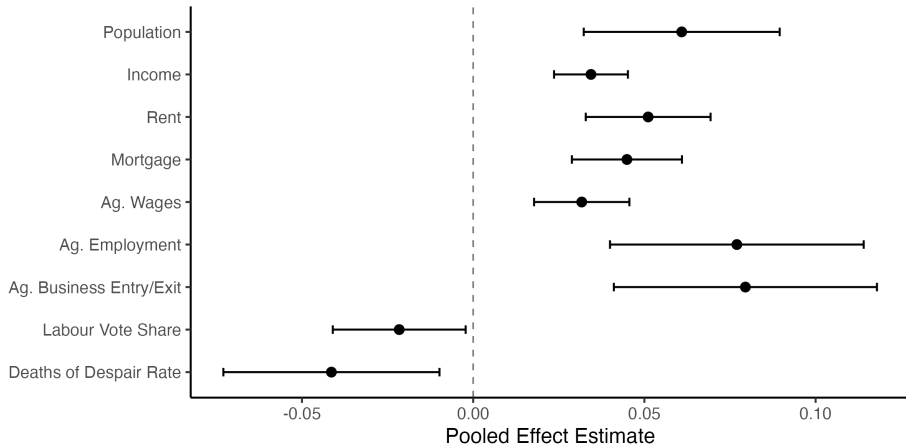


Notes: Effect of permanent increase of 0.0816 ET/hectare on income, rent, and mortgage. Equivalent to a 10% increase in irrigation intensity at the average SA1. $\approx 5,300$ SA1 obs. each year. [TableIncome](#) [TableRent](#) [TableMortgage](#)

Summary of all Local Impacts

[AgMain](#)[PolMain](#)[DeathMain](#)

Summary of all Local Impacts

[AgMain](#)[PolMain](#)[DeathMain](#)

- **Next step:** Do all of these community impacts capitalize into home prices?

What happens to home prices?

Resale IV approach Data

$$\log(p_{l,t}) = \tau_t \Delta \hat{W}_l + \phi_i + \gamma_{d,t} + \varepsilon_{l,t}$$

$p_{i,t}$: Sale price of parcel i in year t

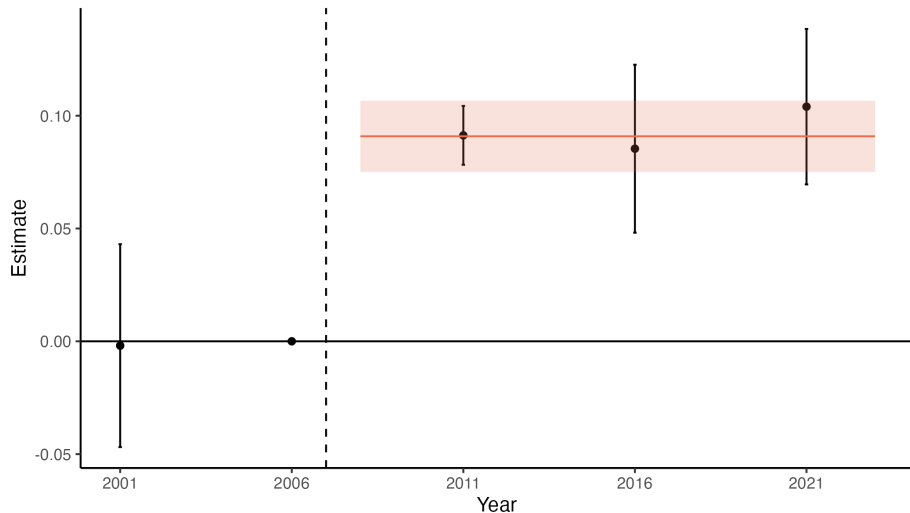
$\Delta \hat{W}_l$: Instrumented average difference in evapotranspiration per hectare

τ_i : Effect of change in water

ϕ_i : Unit fixed effects

$\gamma_{d,t}$: District-by-year fixed effects

What happens to home prices?

[OLSFig](#)[Table](#)

Notes: Effect of permanent increase of 0.0816 ET/hectare on home resales in NSW. Equivalent to a 10% increase in irrigation intensity at the average SA1.

Estimate Communal Value of Water: Residential Choice Model

- Missing (public goods) market for community to pay farmers to keep water.
- How much is a community willing to pay?
- Estimate the preference for water in the community with a residential choice model.
 - Map local water, housing price, income, and migration chance to welfare (in \$)
 - Households have different preference for locations
 - Heterogeneous preference for water & price (by density)
 - Income elasticities
 - Estimate using SA2 $\rightarrow$ SA1 migration flows.

Model of Residential Choice: Utility

- **Model:** Extension of Bayer et al. (2007) with locational substitution patterns.
- **Utility:** Representative i from origin $o(i, t)$ SA2 gets utility from moving to SA1 d in t .

$$u_{i,d,t}^{o(i,t)} = \underbrace{\gamma_d^{o(i,t)}}_{\text{location prefs.}} - \underbrace{\alpha_d \log(p_{j,t})}_{\text{housing price}} + \underbrace{\omega_d W_{d,t}}_{\text{water prefs.}} + \underbrace{\theta \log(I_{d,t})}_{\text{income}} + \underbrace{\eta_{wd(d),t}}_{\text{district-year FE}} + \underbrace{\xi_{d,t}^{o(i)}}_{\text{unobs.}} + \underbrace{\varepsilon_{i,d,t}}_{\text{T1EV}}$$

- **Heterogeneity:** Price and water elasticities vary in pre-intervention density D_d

$$\alpha_d = \alpha_0 + \alpha_1 \log(D_d)$$

$$\omega_d = \omega_0 + \omega_1 \log(D_d)$$

Model of Residential Choice: Utility

- **Model:** Extension of Bayer et al. (2007) with locational substitution patterns.
- **Utility:** Representative i from origin $o(i, t)$ SA2 gets utility from moving to SA1 d in t .

$$u_{i,d,t}^{o(i,t)} = \underbrace{\gamma_d^{o(i,t)}}_{\text{location prefs.}} - \underbrace{\alpha_d \log(p_{j,t})}_{\text{housing price}} + \underbrace{\omega_d W_{d,t}}_{\text{water prefs.}} + \underbrace{\theta \log(I_{d,t})}_{\text{income}} + \underbrace{\eta_{wd(d),t}}_{\text{district-year FE}} + \underbrace{\xi_{d,t}^{o(i)}}_{\text{unobs.}} + \underbrace{\varepsilon_{i,d,t}}_{\text{T1EV}}$$

- **Heterogeneity:** Price and water elasticities vary in pre-intervention density D_d

$$\alpha_d = \alpha_0 + \alpha_1 \log(D_d)$$

$$\omega_d = \omega_0 + \omega_1 \log(D_d)$$

- **Caveat:** Currently, crude management of zero-shares (Gandhi et al. 2023).

Model of Residential Choice: Estimation

- **Strategy:** Berry-style inversion and apply instrumental variables GMM.

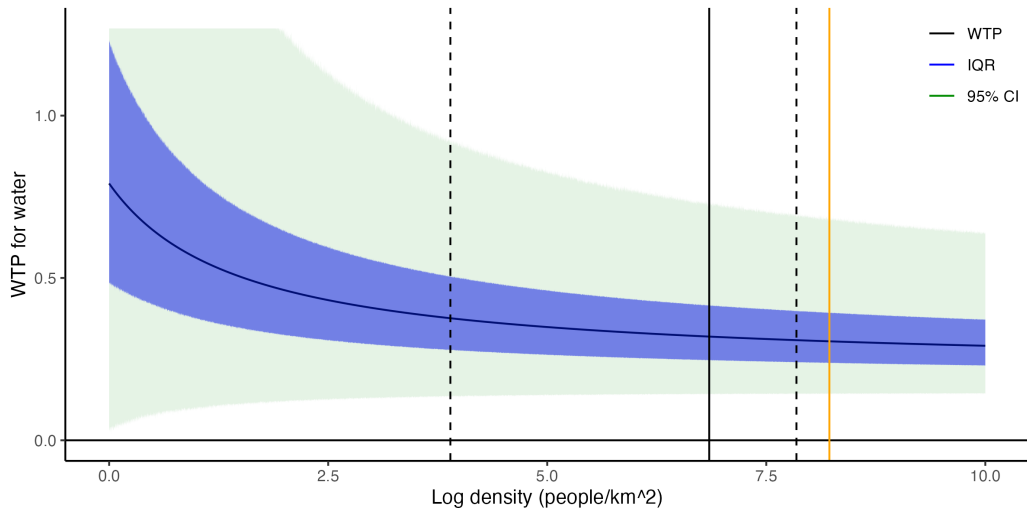
Variable	Instruments	Intuition
Price: $p_{d,t}$	Death count $\times$ prev.period exposure to elderly F-stat: 42.2	<i>Supply Shifter</i>
Water: $W_{d,t}$	Immutable soil chars. $\times$ year F-stat: 204.7	<i>Exog. Productivity</i>
Income: $I_{d,t}$	Bartik instrument F-stat: 60.2	<i>Sectoral wage shifters</i>

- **Data:** Repeated cross-section of SA2 $\rightarrow$ SA1 migration flows form 2006-2021.

Results: Residential demand estimation

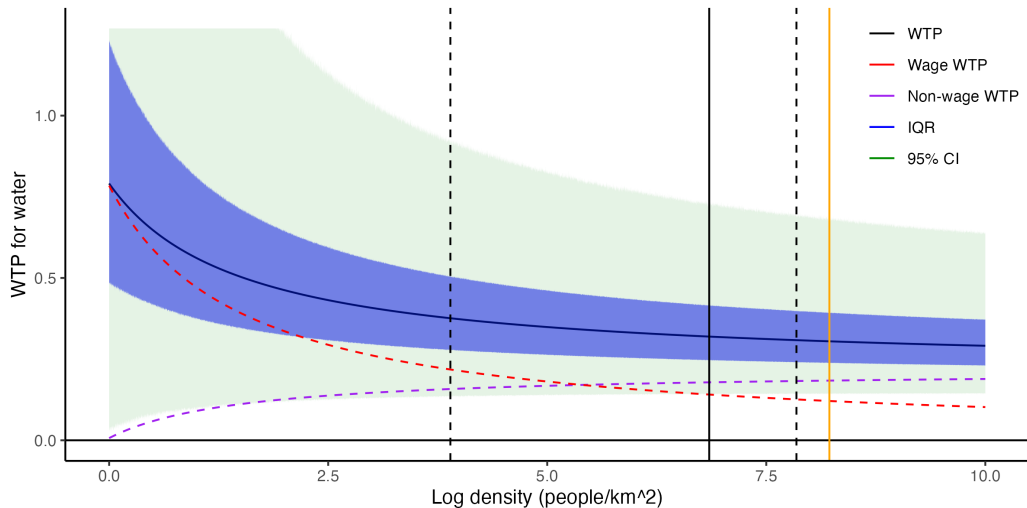
	Main			OLS
Price: α_0	-0.248** (0.110)	-0.207 (0.130)	-0.224 (0.248)	0.039*** (0.007)
Price: α_1	-0.165*** (0.049)	-0.218*** (0.046)	0.133 (0.146)	-0.002 (0.005)
Water: ω_0	0.006 (0.321)	-0.254 (0.337)	-0.360*** (0.086)	-0.362*** (0.078)
Water: ω_1	0.130*** (0.020)	0.096*** (0.021)	0.022* (0.012)	0.020* (0.011)
Wage: θ	0.691*** (0.132)	0.201*** (0.033)	0.176*** (0.057)	0.129*** (0.011)
<i>Instr.: Price</i>	✓	✓	✓	
<i>Instr.: Water</i>	✓	✓		
<i>Instr.: Wage</i>	✓			

Local WTP for Water



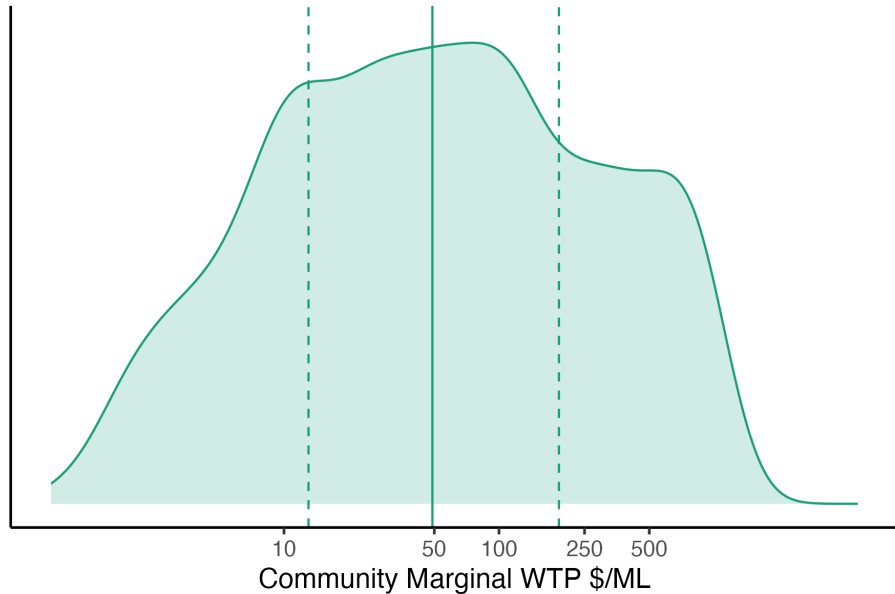
Note: WTP is utility-equivalent percentage home price decrease per 1 SD water increase. Pecuniary refers to the wage effect. Dashed vertical lines are the 25th and 75th percentile of locations, the black line is the mean, and the orange line is the median resident's location. Inference using Gaussian multiplier bootstrap.

Local WTP for Water



Note: WTP is utility-equivalent percentage home price decrease per 1 SD water increase. Pecuniary refers to the wage effect. Dashed vertical lines are the 25th and 75th percentile of locations, the black line is the mean, and the orange line is the median resident's location. Inference using Gaussian multiplier bootstrap.

Estimated Community Marginal WTP for Water



Roadmap

1. Water in Australia
2. Communal Value of Water: Residential Choice
3. Market Value of Water: Agricultural Production
4. Market Design and Counterfactuals

Agricultural WTP for Water

- Need to understand value of productive reallocation.
- **For now:** Estimate simple model using Census of Agriculture from 2005-2020.
- **Production Function:** Cobb-Douglas for crop types $s \in \{\text{annual, dairy, perennial}\}$.

$$\underbrace{Y_{l,t,s}}_{\text{output (\$)}} = \underbrace{\exp(\omega_{l,t,s})}_{\text{productivity}} \underbrace{W_{l,t,s}^{\alpha_W^s}}_{\text{water}} \underbrace{L_{l,t,s}^{\alpha_L^s}}_{\text{labor}}$$

- Can back out marginal WTP from parameters.
- **For future:** Serious treatment of functional form, productivity, and inframarginal WTP.

Estimate Agricultural Production

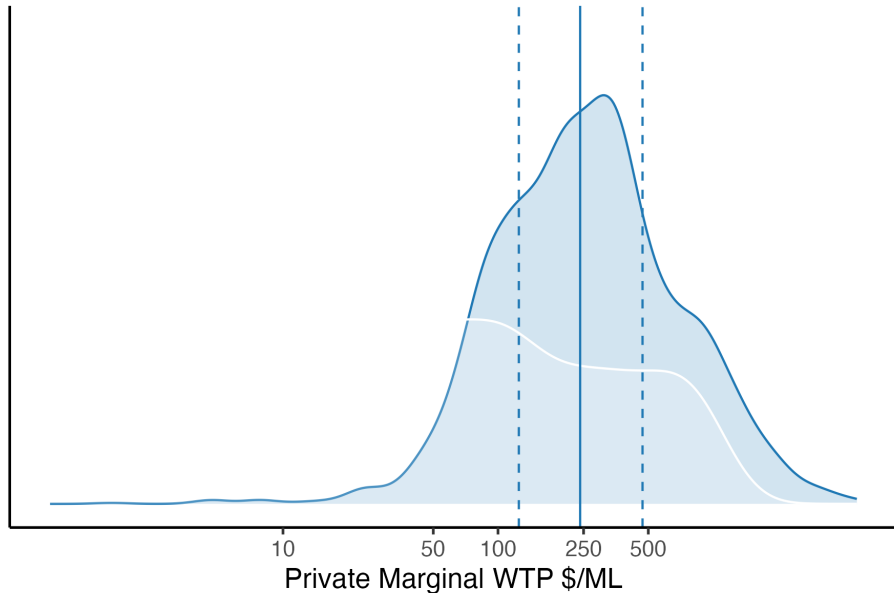
- **Strategy:** Instrument for changes in water and labor from 2005-2020.

$$\underbrace{\Delta \log(Y_{l,s})}_{\text{output (\$)}} = \underbrace{\alpha_W^s \Delta \log(W_{l,s})}_{\text{water}} + \underbrace{\alpha_L^s \Delta \log(L_{l,s})}_{\text{labor}} + \underbrace{\phi_{d(l),s}}_{\text{district fe}} + \varepsilon_{l,s}$$

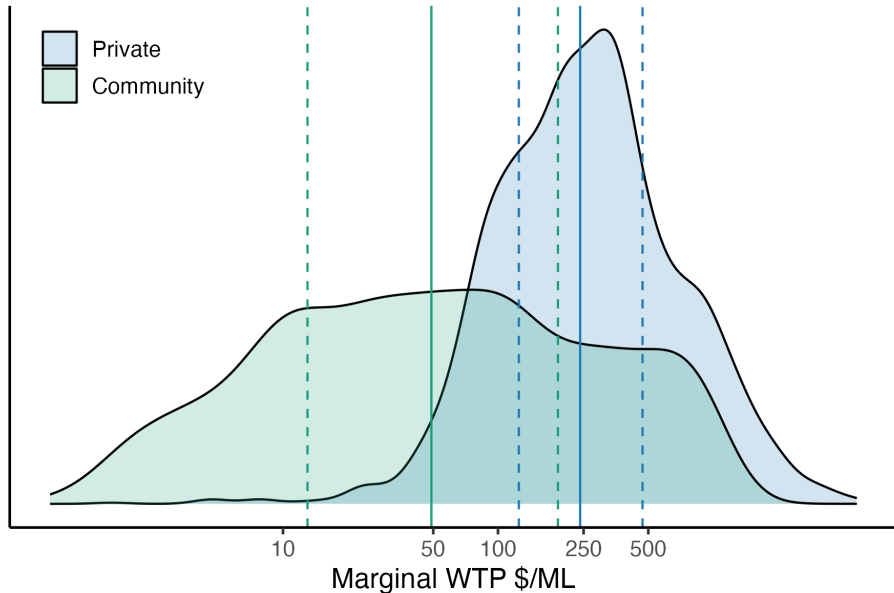
- **Water instrument:** Rain in SA2.
- **Labor instrument:** 2005 population share in SA2 working in *other* agricultural sectors.

Sector	α_W	α_L
Annual	0.248*** (0.053)	0.302* (0.176)
Dairy	0.359*** (0.100)	0.296*** (0.108)
Perennial	0.842*** (0.122)	0.377 (0.264)

Estimated Agricultural Marginal WTP for Water



Estimated Agricultural Marginal WTP for Water



Correlation between trade, agricultural value, and local WTP

- Median farmer WTP of **\$247/ML** and community WTP of **\$49/ML**.
- Predict post-2007 change in water with SA1 farmer WTP and community WTP.

$$\Delta W_l = \underbrace{\beta_F \log(P_l^{MWTP})}_{\text{private wtp}} + \underbrace{\beta_C \log(C_l^{MWTP})}_{\text{local wtp}} + \underbrace{\phi_{d(l)} + \eta_{SA2(l)}}_{\text{regional FEs}} + \varepsilon_l$$

- Private and community incentives are negatively correlated.

Dependent Var.:	$\frac{W^{Post}}{W^{Pre}}$ OLS	$W^{Post} - W^{Pre} > 0$ Logit	log(Community WTP)
log(Private WTP)	0.008 (0.002)	0.339 (0.014)	-0.051 (0.009)
log(Community WTP)	-0.002 (0.002)	-0.263 (0.091)	

Roadmap

1. Water in Australia
2. Communal Value of Water: Residential Choice
3. Market Value of Water: Agricultural Production
4. Market Design and Counterfactuals

Framework: Policy? Market Design? Nothing?

- **Efficiency:**
- **Equity:**
- **Political Feasibility:**

Framework: Policy? Market Design? Nothing?

- **Efficiency:**

- Let the market do its thing (Arrow and Debreu 1954, Coase 1960, Stavins 1995).
- **But**, if private WTP is negatively correlated with community WTP and sufficient heterogeneity, scope for improving efficiency (Slattery 2025, Kline and Moretti 2014).

- **Equity:**

- **Political Feasibility:**

Framework: Policy? Market Design? Nothing?

- **Efficiency:**

- Let the market do its thing (Arrow and Debreu 1954, Coase 1960, Stavins 1995).
- **But**, if private WTP is negatively correlated with community WTP and sufficient heterogeneity, scope for improving efficiency (Slattery 2025, Kline and Moretti 2014).

- **Equity:**

- Let the market do its thing and redistribute (Atkinson and Stiglitz 1976).
- **But**, if WTP provides more information about welfare weights or marginal utility of income, distortions may be desired in goods markets (Doligalski et al. 2025).

- **Political Feasibility:**

Framework: Policy? Market Design? Nothing?

- **Efficiency:**

- Let the market do its thing (Arrow and Debreu 1954, Coase 1960, Stavins 1995).
- **But**, if private WTP is negatively correlated with community WTP and sufficient heterogeneity, scope for improving efficiency (Slattery 2025, Kline and Moretti 2014).

- **Equity:**

- Let the market do its thing and redistribute (Atkinson and Stiglitz 1976).
- **But**, if WTP provides more information about welfare weights or marginal utility of income, distortions may be desired in goods markets (Doligalski et al. 2025).

- **Political Feasibility:**

- People care about equity.
- Either don't have tax lever OR indirect redistribution is not salient to impacted parties.
- Mechanisms that generate **Pareto-improving outcomes** would be ideal.

Framework: Policy? Market Design? Nothing?

- **Efficiency:**

- Let the market do its thing (Arrow and Debreu 1954, Coase 1960, Stavins 1995).
- **But**, if private WTP is negatively correlated with community WTP and sufficient heterogeneity, scope for improving efficiency (Slattery 2025, Kline and Moretti 2014).

- **Equity:**

- Let the market do its thing and redistribute (Atkinson and Stiglitz 1976).
- **But**, if WTP provides more information about welfare weights or marginal utility of income, distortions may be desired in goods markets (Doligalski et al. 2025).

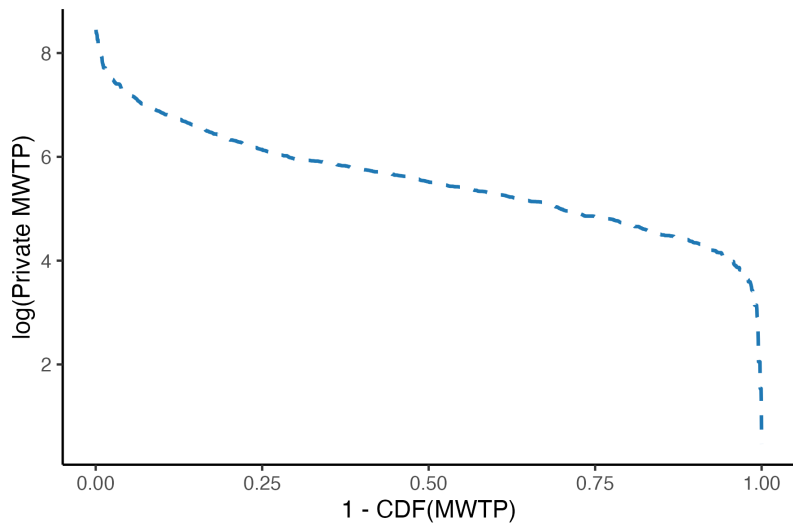
- **Political Feasibility:**

- People care about equity.
- Either don't have tax lever OR indirect redistribution is not salient to impacted parties.
- Mechanisms that generate **Pareto-improving outcomes** would be ideal.

- **Today:** Explore Pareto gains given joint distribution of private and social WTP.

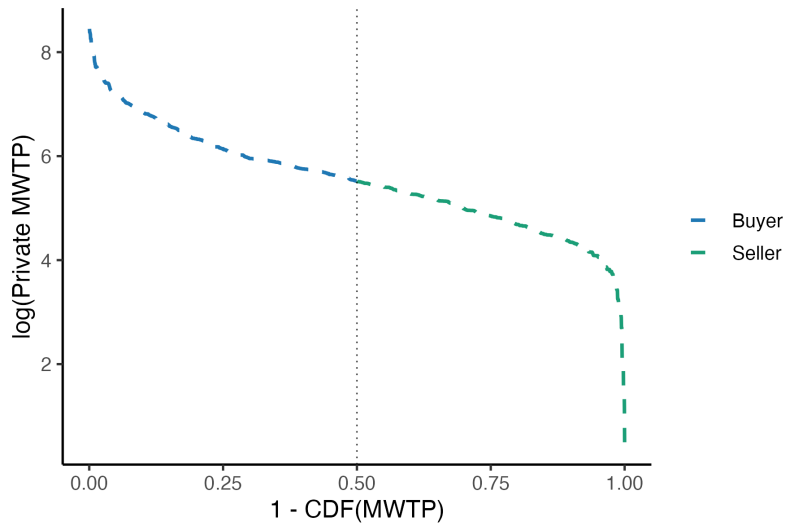
Efficiency and Private Gains

- Distribution of Private MWTP for water



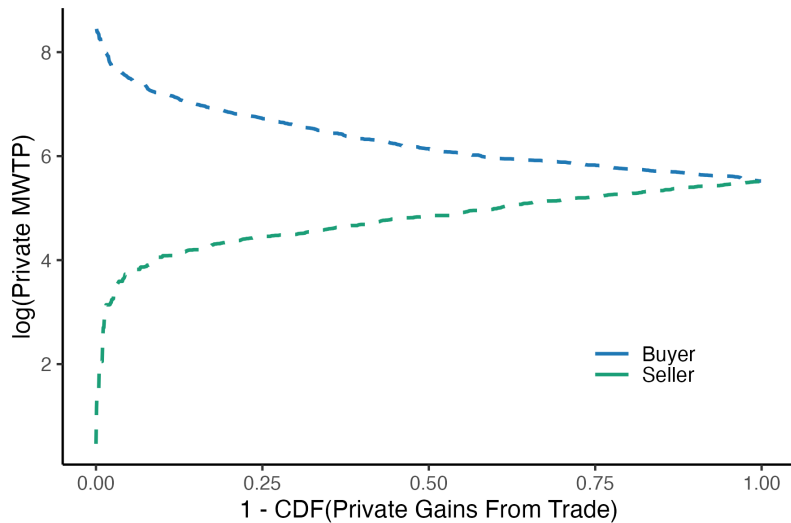
Efficiency and Private Gains

- Distribution of Private MWTP for water
- Identify buyers and sellers



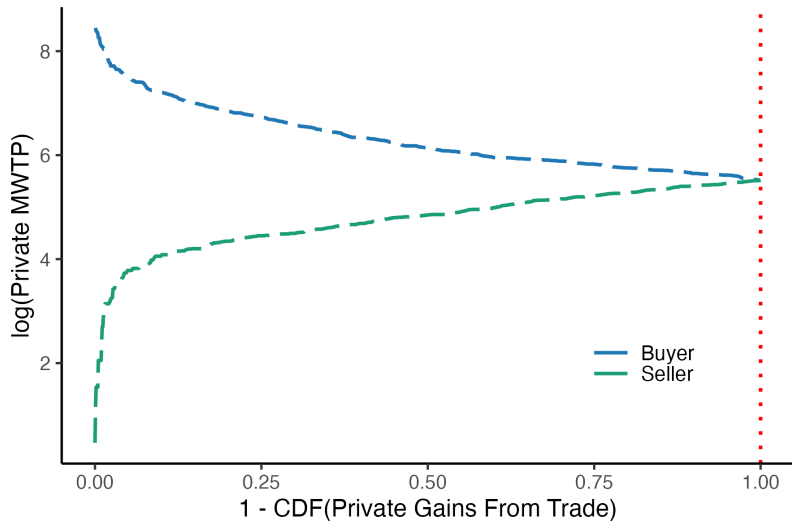
Efficiency and Private Gains

- Distribution of Private MWTP for water
- Identify buyers and sellers
- Order pairs by private gains from trade



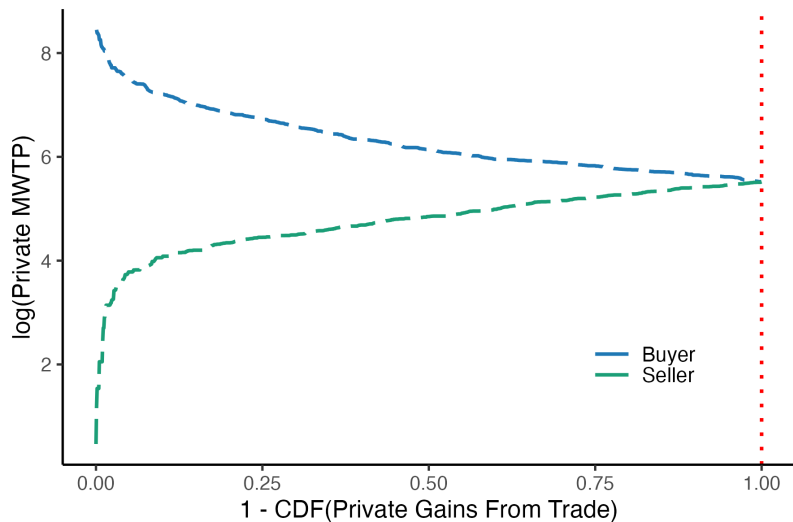
Efficiency and Private Gains

- Distribution of Private MWTP for water
- Identify buyers and sellers
- Order pairs by private gains from trade
- Achieve 100% of private gainful trades



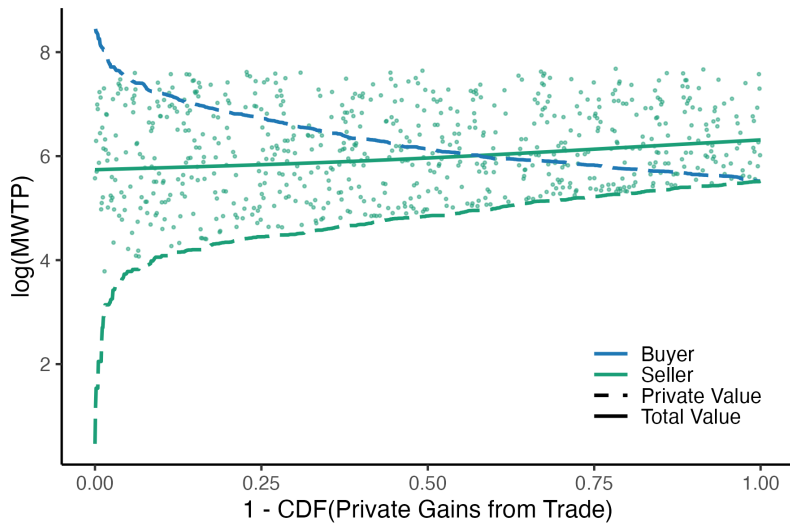
Status quo thinking: Private buyer is responsible for damages

- Keep this pairing of buyers and sellers



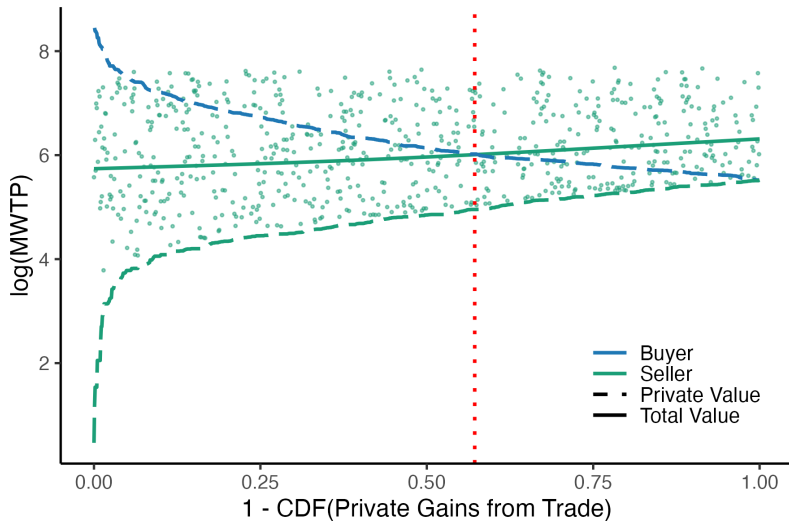
Status quo thinking: Private buyer is responsible for damages

- Keep this pairing of buyers and sellers
- Incorporate seller community value
- Private buyer responsible for transfer to community



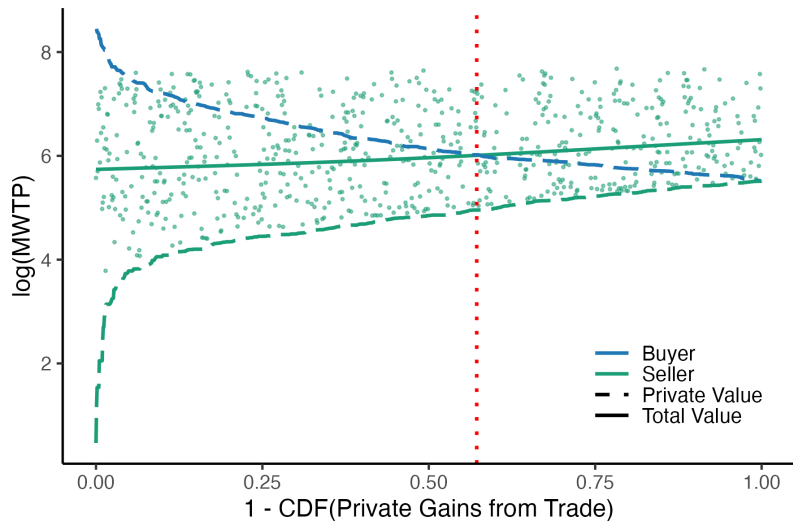
Status quo thinking: Private buyer is responsible for damages

- Keep this pairing of buyers and sellers
- Incorporate seller community value
- Private buyer responsible for transfer to community
- 57% of gainful trades



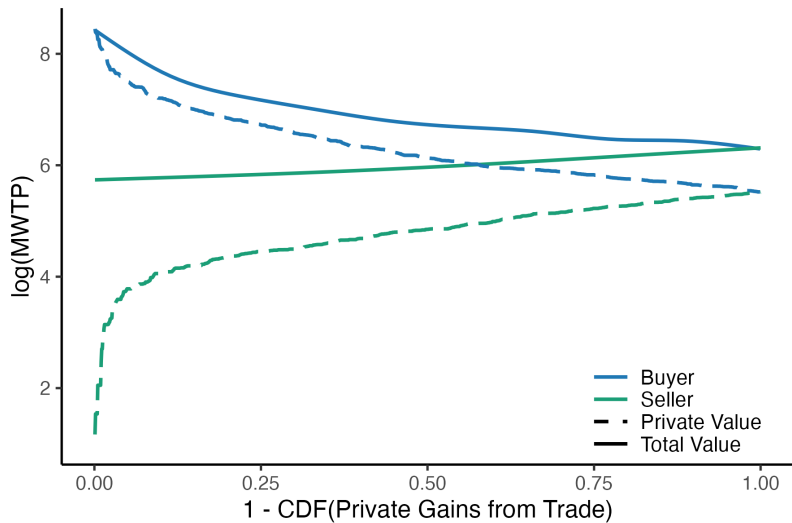
Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits



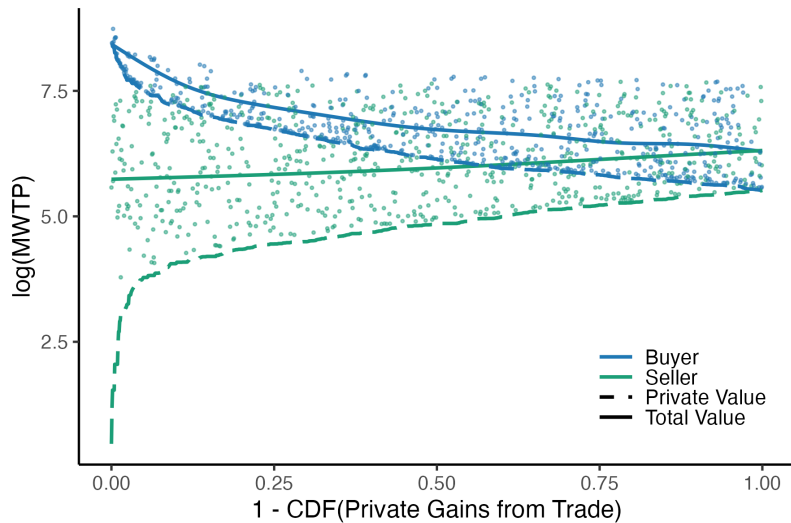
Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits
- Including buying community shows more gains



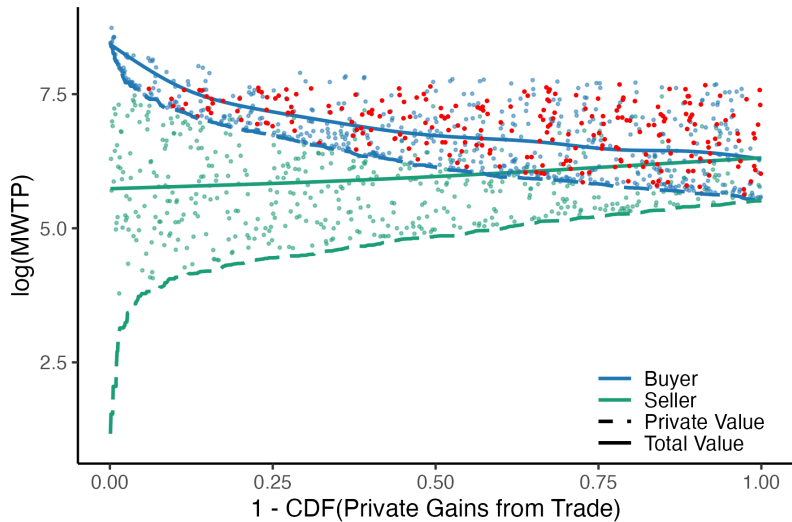
Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits
- Including buying community shows more gains
- Plot masks heterogeneity



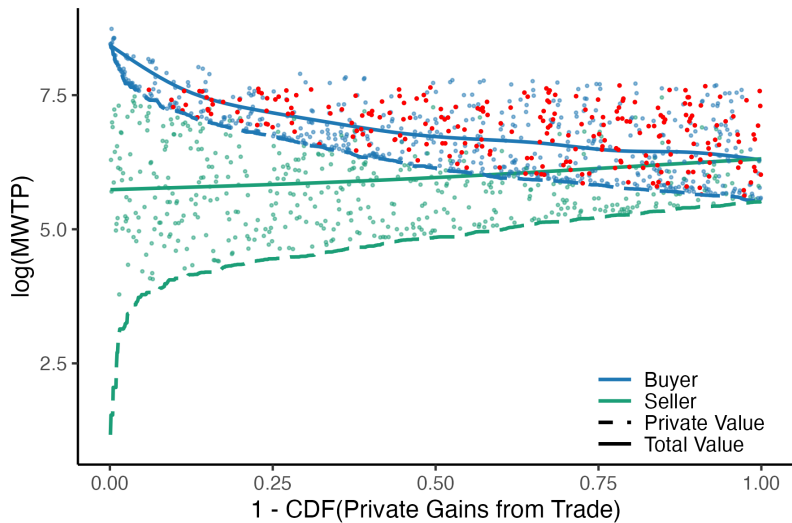
Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits
- Including buying community shows more gains
- Plot masks heterogeneity
- 21% are myopically inefficient



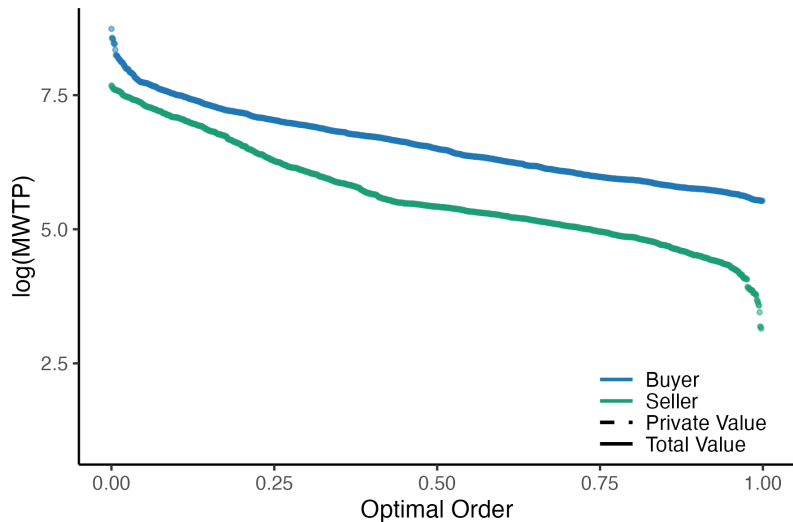
Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits
- Including buying community shows more gains
- Plot masks heterogeneity
- 21% are myopically inefficient
- Matching buyers/sellers



Economist Frustration: Buyer's community benefits not considered

- Myopic consideration of costs/benefits
- Including buying community shows more gains
- Plot masks heterogeneity
- 21% are myopically inefficient
- Matching buyers/sellers



Conclusion and Next Steps

- Estimate significant local spillovers from water trade in MDB.
- Median local WTP for water around 20% of private WTP.
- Achieving widespread Pareto transfers requires incorporation of buying community WTP.

Conclusion and Next Steps

- Estimate significant local spillovers from water trade in MDB.
- Median local WTP for water around 20% of private WTP.
- Achieving widespread Pareto transfers requires incorporation of buying community WTP.
- **Next Steps:**
 - Are people buying the event-study design?
 - How do we want to improve production model?
 - How hard should we pursue actual trade data?
 - How GE do we want to be?
 - Where do we want to sit on the efficiency, equity, vs. political design question?

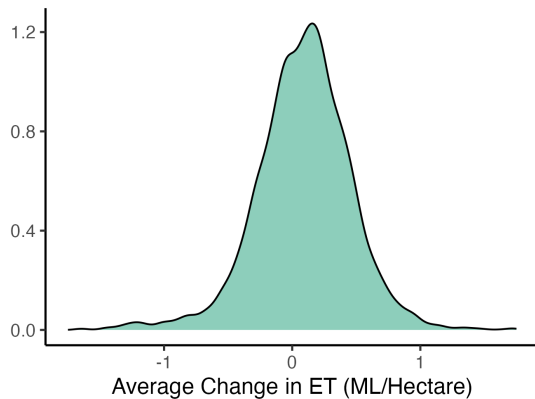
Thanks!

Appendix

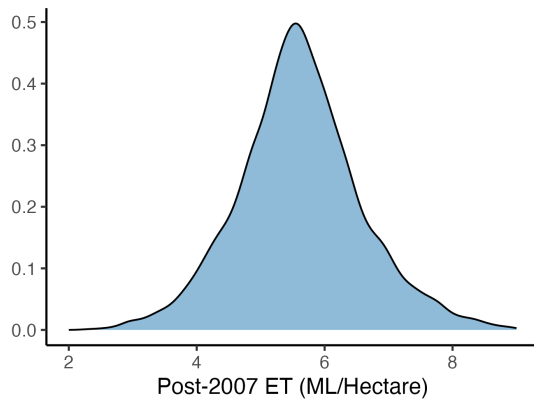
Evapotranspiration Distribution

[Back](#)

A. Change in evapotranspiration

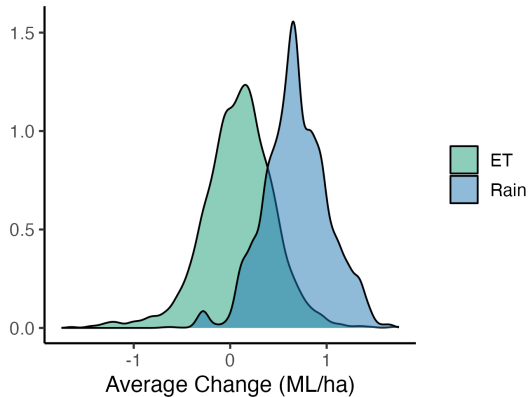


B. Post-2007 evapotranspiration



Evapotranspiration Distribution [Back](#)

A. Change in evapotranspiration and rain



B. Evapotranspiration change vs. Rain change
ET Change vs Rainfall Change

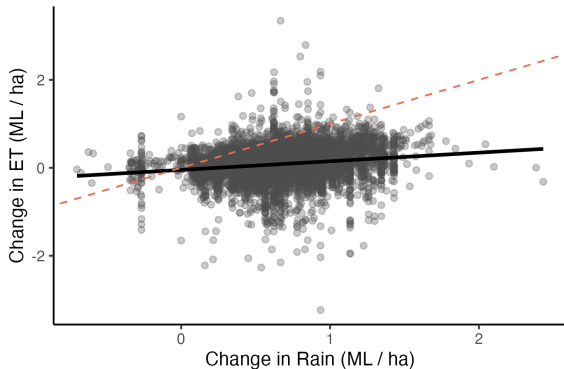


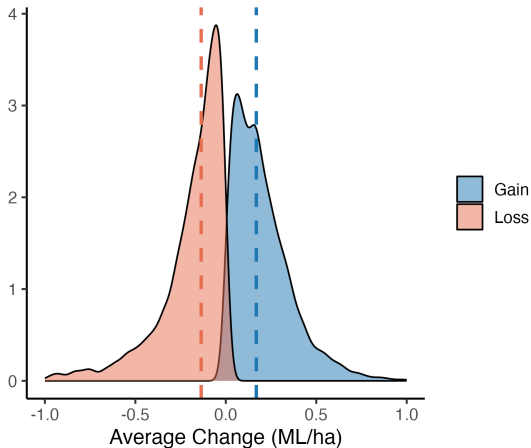
Table: Changes in population by change in ET (post 2007)

Dependent Variables: Model:	Total Resid ET (1)	(2)	Dry-Month Resid ET (3)	(4)	Wet-Month Resid ET (5)	(6)
<i>Variables</i>						
Allocation Volume	0.7193 (0.4188)	1.613** (0.5929)	1.204*** (0.3451)	1.211*** (0.3411)	-0.1248 (0.1243)	0.0322 (0.1514)
Net Trade		2.642 (1.976)		1.985* (1.068)		0.2148 (0.3323)
<i>Fixed-effects</i> region_id	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	440	210	440	210	440	210
R ²	0.01414	0.09338	0.54999	0.70481	0.85546	0.88487
Within R ²	0.00890	0.04566	0.05616	0.06759	0.00358	0.00054

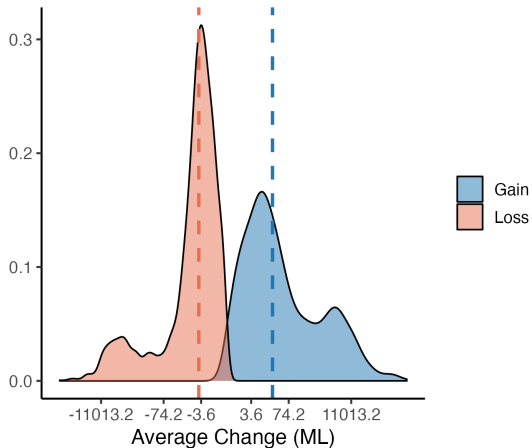
Clustered (region_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Residualized Evapotranspiration Distribution [Back](#)

A. Change in residualized evapotranspiration/ha

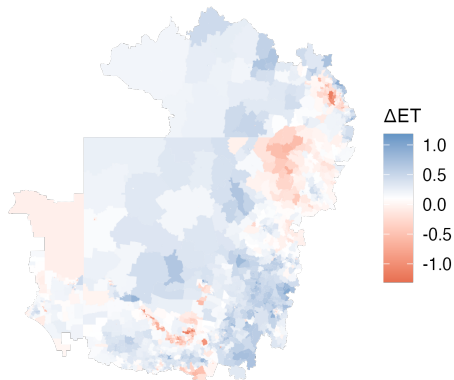


B. Change in residualized evapotranspiration/ha

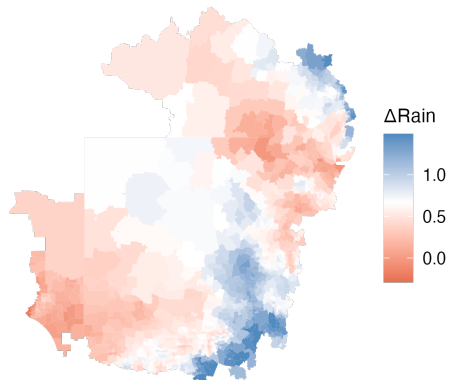


Evapotranspiration Geography [Back](#)

A. Change in evapotranspiration (ML/ha)

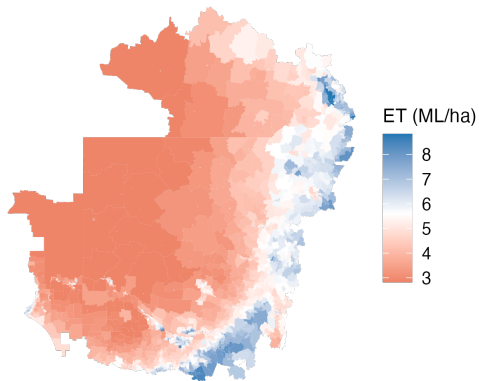


B. Change in rain (ML/ha)

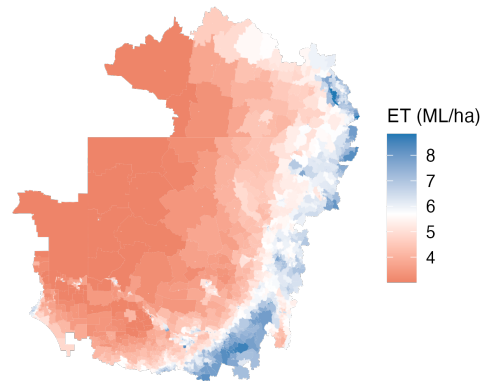


Note: Evapotranspiration data from satellite images. Rain data from CHIRPS.

A. Pre-2007 evapotranspiration (ML/ha)



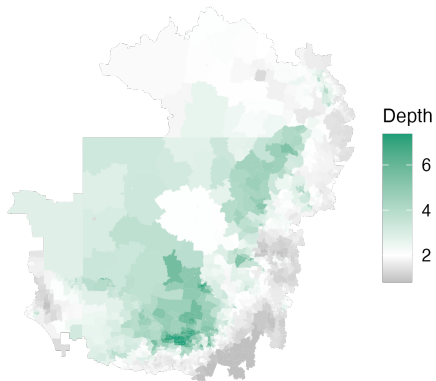
B. Post-2007 evapotranspiration (ML/ha)



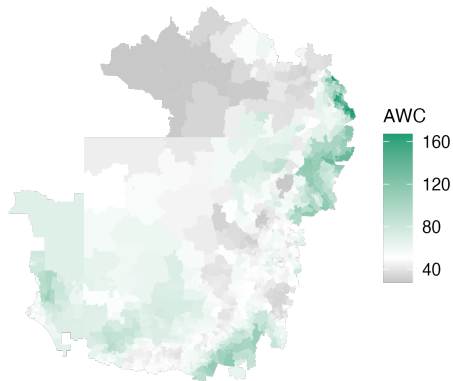
Note: Evapotranspiration data from satellite images.

Soil Characteristic Geography [Back](#)

A. Average soil depth



B. Average soil available water capacity



Note: Average by SA1. From the Soil and Landscape Grid of Australia.

Table: First-Stage IV

Dependent Variable: Model:	Residual ET Change (ML/ha) (1)
<i>Variables</i>	
AWC	0.0204** (0.0099)
Depth	0.6265*** (0.2396)
pH	-5.339* (2.730)
pH ²	0.4513* (0.2349)
AWC × Depth	-0.0134*** (0.0040)
<i>Fixed-effects</i>	
water_id	Yes
<i>Fit statistics</i>	
Observations	6,550
R ²	0.11679

Clustered (SA2_CODE21) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table: Changes in income by change in ET (post 2007)

Dependent Variables: Model:	log(inc_96) (1)	log(inc_01) (2)	log(inc_11) (3)	log(inc_16) (4)	log(inc_21) (5)	pooled (6)
<i>Variables</i>						
change_per_land	-0.0171** (0.0078)	-0.0026 (0.0081)	0.0402*** (0.0110)	0.0230*** (0.0080)	0.0403*** (0.0110)	0.0344*** (0.0055)
<i>Fixed-effects</i>						
water_id	Yes	Yes	Yes	Yes	Yes	
year-water_id						Yes
<i>Fit statistics</i>						
Observations	5,315	5,454	5,029	5,571	5,627	16,227
R ²	0.03434	0.02274	0.02383	0.04141	-0.09833	0.24986
Within R ²	-0.00381	-0.00737	-0.11842	-0.02734	-0.16087	-0.09867
F-test (1st stage)	43.423	43.037	36.367	39.400	37.480	112.95

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table: Changes in rent by change in ET (post 2007)

Dependent Variables: Model:	log(rent_96) (1)	log(rent_01) (2)	log(rent_11) (3)	log(rent_16) (4)	log(rent_21) (5)	pooled (6)
<i>Variables</i>						
change_per_land	0.0212 (0.0178)	0.0164 (0.0211)	0.0334** (0.0162)	0.0471*** (0.0151)	0.0688*** (0.0231)	0.0511*** (0.0093)
<i>Fixed-effects</i>						
water_id	Yes	Yes	Yes	Yes	Yes	
year-water_id						Yes
<i>Fit statistics</i>						
Observations	5,338	5,435	4,674	5,251	5,551	15,476
R ²	0.12767	0.13205	-0.04109	-0.11048	-0.14555	0.08896
Within R ²	-0.02843	-0.00687	-0.05450	-0.14600	-0.23765	-0.14094
F-test (1st stage)	42.763	43.019	33.755	38.176	39.554	111.36

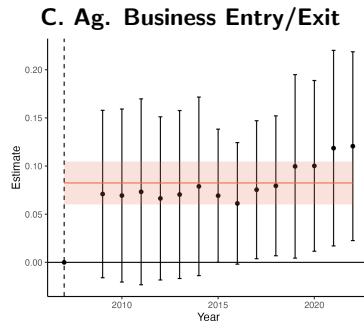
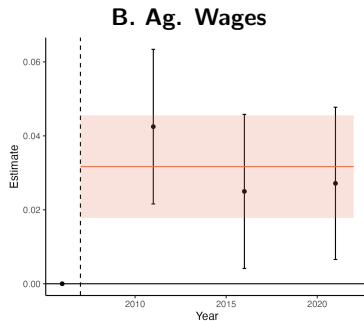
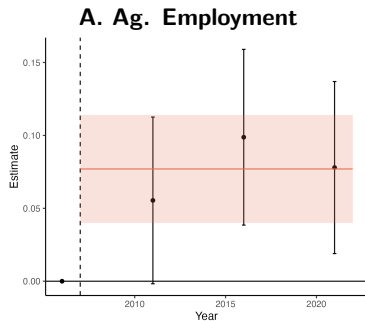
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table: Changes in mortgage by change in ET (post 2007)

Dependent Variables: $\log(\text{mtg_96}) - \log(\text{mtg_06})$	$\log(\text{mtg_01}) - \log(\text{mtg_06})$	$\log(\text{mtg_11}) - \log(\text{mtg_06})$	$\log(\text{mtg_16}) - \log(\text{mtg_06})$	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
change_per_land	-0.0146 (0.0163)	-0.0192 (0.0163)	0.0342** (0.0132)	0.0422*** (0.0132)
<i>Fixed-effects</i>				
water_id	Yes	Yes	Yes	Yes
year-water_id				
<i>Fit statistics</i>				
Observations	5,291	5,419	4,931	5,442
R ²	0.15343	0.12554	0.00627	-0.07408
Within R ²	-0.01340	-0.02485	-0.04928	-0.13657
F-test (1st stage)	42.833	42.894	37.293	40.584

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

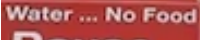
Agricultural Labor Market Effects

[GoBack](#)

Note: Effect of permanent increase in ET/hectare on agricultural employment, wages, and entry/exit. $\approx 4,300$ SA1 obs. each year.

[AgJob](#)[AgWage](#)

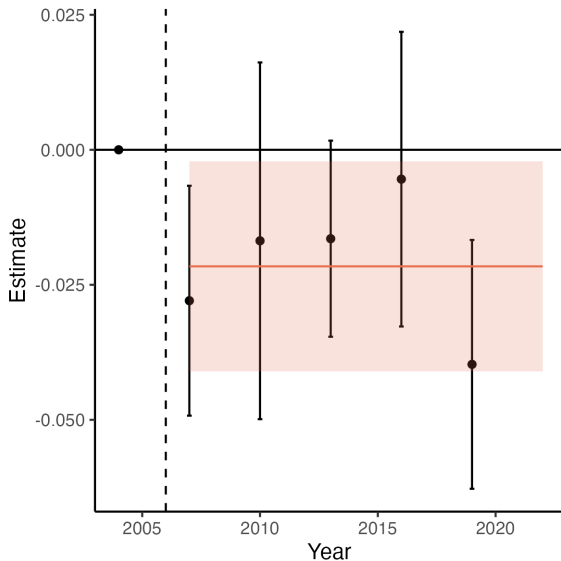
GoBack



Water losers shift politically

[GoBack](#)

- **Labor:** Campaign on MDB plan regulation and overhaul
- **Liberal-National:** “Continue to advance the Plan as it stands”
- Outcome is $\text{Log}(\text{Labor Vote Share})$
- Polling data $\text{SA2} \times \text{year}$ (2004-2019)

[Table](#)

Water fights

Much of the world is desperately short of fresh water. Are future water wars inevitable?

BY NANCY MACDONALD JULY 6, 2009

Every few days, another farmer commits suicide in Australia's Murray-Darling Basin, the agricultural heartland. Many, according to Australian evolutionary biologist Tim Flannery, haven't had any water in almost four years—in places, the allocation of irrigation water has been cut to zero. Their farms have dried up, leaving a dusty, wind-whipped scrubland. Cattle bellow from hunger through the night. "Despair is an enormous problem," says Flannery. "There is no sign the situation will ever improve." Government has compiled a suicide watch list.

Deaths of Despair [GoBack](#)

- Suicide from decline of farming towns
- Deaths of despair rate ($SA2 \times year$)
- Years: 03-07, 08-12, 10-14, 11-15, 14-18

$$\log(D_{lt}) - \log(D_{l,2005})$$

- Noisy, but suggestive evidence.

[Table](#)

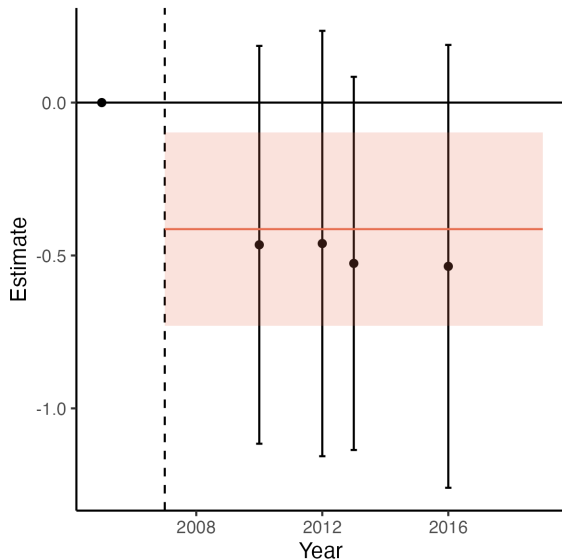


Table: Changes in Ag. Wages by change in ET (post 2007)

Dependent Variables: log(ag_income_2011) log(ag_income_2016) log(ag_income_2021) pooled	(1)	(2)	(3)	(4)
Model:				
<i>Variables</i>				
change_per_land	0.0425*** (0.0107)	0.0250** (0.0106)	0.0272** (0.0105)	0.0317*** (0.0071)
<i>Fixed-effects</i>				
water_id	Yes	Yes	Yes	Yes
water_id-year				
<i>Fit statistics</i>				
Observations	2,581	2,506	2,563	7,650
R ²	-0.06656	0.00440	0.00441	0.12673
Within R ²	-0.08138	-0.01716	-0.01897	-0.03437
F-test (1st stage)	43.159	42.546	46.693	132.38

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table: Changes in Ag. Jobs by change in ET (post 2007)

Dependent Variables: log(total_ag_jobs) Model:	(1)	(2)	(3)	pooled (4)
<i>Variables</i> change_per_land	0.0554* (0.0292)	0.0988*** (0.0307)	0.0779** (0.0301)	0.0770*** (0.0189)
<i>Fixed-effects</i> water_id water_id-year	Yes	Yes	Yes	Yes
<i>Fit statistics</i> Observations	2,593	2,514	2,568	7,675
R ²	0.02071	-0.07097	-0.04561	-0.02730
Within R ²	-0.03896	-0.11911	-0.07644	-0.07411
F-test (1st stage)	43.441	42.746	46.732	132.93

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table: Changes in ALP VoteShare by change in ET (post reform)

Dependent Variables: Model:	log(alp_share_07) (1)	log(alp_share_10) (2)	log(alp_share_13) (3)	log(alp_share_16) (4)	log(alp_share_19) (5)	pooled (6)
<i>Variables</i>						
change_per_land	-0.0279** (0.0109)	-0.0169 (0.0114)	-0.0165* (0.0093)	-0.0054 (0.0139)	-0.0397*** (0.0118)	-0.0216** (0.0099)
<i>Fixed-effects</i>						
water_id	Yes	Yes	Yes	Yes	Yes	
year-water_id						Yes
<i>Fit statistics</i>						
Observations	262	225	226	250	258	1,221
R ²	0.29617	0.42745	0.39070	0.33251	0.28004	0.40583
Within R ²	-0.26564	-0.10025	-0.04961	-0.01376	-0.24149	-0.10332
F-test (1st stage)	12.402	13.538	13.537	12.298	11.510	57.169

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table: Population Event-Study [Back](#)

Table: Changes in population by change in ET (post 2007)

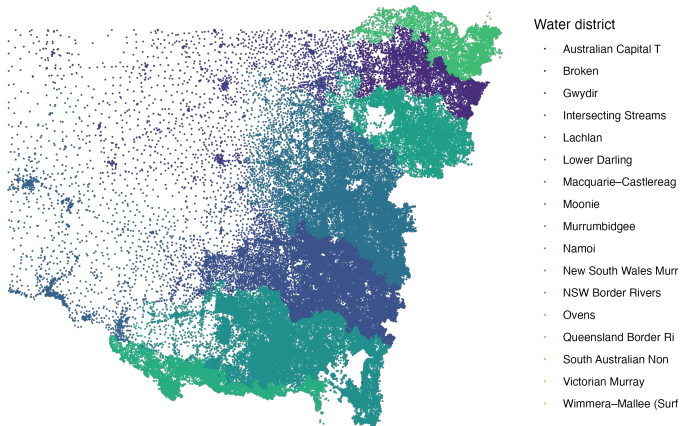
Dependent Variables: Model:	log(pop_1991) (1)	log(pop_1996) (2)	log(pop2001) (3)	log(pop_2011) (4)	log(pop_2016) (5)	log(pop_2021) (6)	pooled (7)
<i>Variables</i>							
change_per_land	-0.0114 (0.0188)	-0.0113 (0.0089)	-0.0078 (0.0348)	0.0429 (0.0271)	0.0563** (0.0260)	0.0814*** (0.0279)	0.0609*** (0.0146)
<i>Fixed-effects</i>							
water_id	Yes	Yes	Yes	Yes	Yes	Yes	
year-water_id							Yes
<i>Fit statistics</i>							
Observations	4,839	5,340	5,452	4,992	5,538	5,548	16,078
R ²	0.02655	0.03627	0.05455	0.00566	-0.05056	-0.15697	-0.06463
Within R ²	0.00438	0.01222	0.00585	-0.07061	-0.10900	-0.20556	-0.12561
F-test (1st stage)	42.746	42.769	43.305	37.309	40.372	39.607	117.09

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table: Changes in suicide rate by change in ET (post 2007)

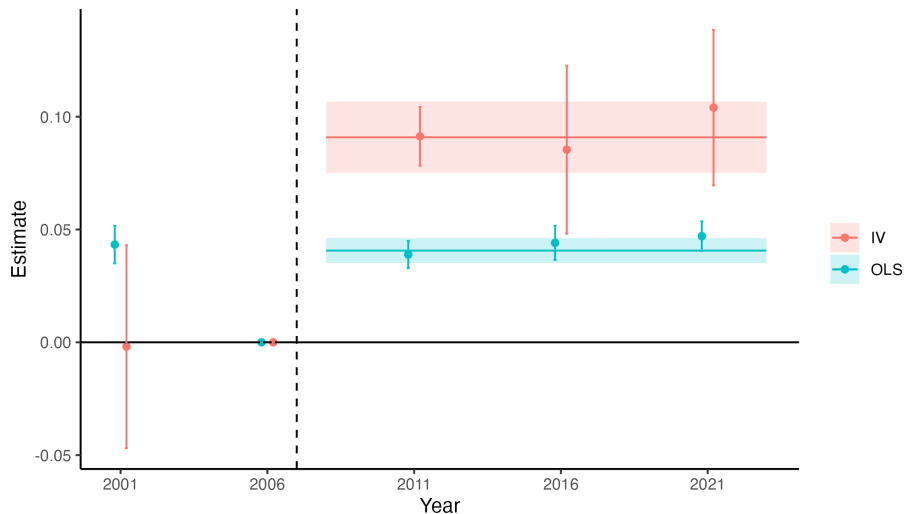
Dependent Variables: Model:	log(suicide_0812) (1)	log(suicide_1014) (2)	log(suicide_1115) (3)	log(suicide_1418) (4)	pooled (5)
<i>Variables</i>					
change_per_land	-0.04651 (0.03319)	-0.04609 (0.03549)	-0.05259 (0.03113)	-0.05355 (0.03694)	-0.04138** (0.01613)
<i>Fixed-effects</i>					
water_id	Yes	Yes	Yes	Yes	Yes
year-water_id					
<i>Fit statistics</i>					
Observations	911	890	890	911	3,602
R ²	-0.17152	-0.13024	-0.18487	-0.13487	-0.03661
Within R ²	-0.50621	-0.44935	-0.52271	-0.61343	-0.37078
F-test (1st stage)	29.966	48.388	48.388	29.966	112.80

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1



Notes: 1,290,017 transactions (with price) in MDB portion of New South Wales from 1995-2024. This contains 1/3 of the total population of MDB as of 2006.

Home Prices Event-Study: IV and OLS Figure [Back](#)



Notes: Effect of permanent increase in ET/hectare on home resales in NSW.

Table: Long run changes in property sale prices

Dependent Variable: Model:	Pooled IV	Pooled OLS	Placebo	^y 2001	2011	2016	2021
<i>Variables</i> change_per_land	0.7633*** (0.1244)	0.3694*** (0.0243)	-0.1004 (0.2220)		0.4029*** (0.1260)	0.7305*** (0.1687)	0.9012*** (0.1536)
<i>Fixed-effects</i> area_type	Yes	Yes			Yes	Yes	Yes
water_id-year_bunch	Yes	Yes	Yes		Yes	Yes	Yes
property_id	Yes	Yes	Yes		Yes	Yes	Yes
<i>Fit statistics</i> Observations	392,337	392,337	142,019		213,015	213,275	221,586
R ²	0.82926	0.83083	0.88381		0.86482	0.87122	0.88303
Within R ²	0.00063	0.00983	-0.00197		0.00882	0.00304	-0.00060
F-test (1st stage)	1,696.5		828.54		821.38	1,029.5	1,251.6

Clustered (SA1_CODE21) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1